

of the risk being optimized. Whereas risk-sensitive RL typically defines risk over return distributions, we define epistemic and normative risk over belief states and dialogue trajectories. This allows risk sensitivity to govern intervention behavior such as clarification, refusal, and self-correction, rather than only the variability of task returns.

Across these literatures, prior work has addressed abstention, clarification, self-correction, and risk sensitivity largely in isolation. Frictive Policy Optimization unifies these threads by treating epistemic interventions as control actions optimized under a single risk-sensitive objective, providing both a formal foundation and a family of practical learning algorithms for epistemically aligned behavior.

3 Frictive Interventions in Dialogue

This section introduces a taxonomy of *frictive interventions* in dialogue. The taxonomy serves two roles. First, it provides a principled conceptual account of the kinds of epistemic resistance that arise naturally in human collaborative interaction. Second, it defines the *action space* over which Frictive Policy Optimization operates. Unlike surface-level stylistic variation, frictive interventions are qualitatively distinct communicative acts that regulate commitment, manage uncertainty, and control epistemic risk.

3.1 From Dialogue Acts to Epistemic Interventions

Dialogue research has long recognized distinctions among communicative acts, such as questions, assertions, requests, and acknowledgments. While these distinctions are essential for modeling conversational structure, they are insufficient for alignment in large language models. Dialogue acts primarily categorize utterances by their conversational role; they do not capture the *epistemic motivation* behind resisting or deferring task completion.

Frictive interventions cut across traditional dialogue act categories. For example, a clarifying question and a rhetorical challenge may both be interrogatives, yet they differ fundamentally in their epistemic function. Similarly, a refusal may take the syntactic form of an assertion, but its role is not to convey information, but to block an action.

We therefore distinguish frictive interventions from dialogue acts in the following sense:

A frictive intervention is defined not by

its surface form, but by its role in regulating epistemic commitment and risk within an interaction.

This formulation resonates with formal models of shared knowledge and commitment in dialogue (Khebour et al., 2024; Obiso et al., 2025), in which interaction state encodes not only surface history but agents’ evolving epistemic commitments and mutual beliefs.

This distinction is also crucial for alignment: a model that merely learns dialogue act frequencies cannot reliably learn when to clarify, challenge, or refuse. What is required is a taxonomy grounded in epistemic function. Furthermore, this observation is supported by recent findings that current language models often fail at precisely this kind of interactional timing, even when they generate locally plausible responses; for instance, models struggle to predict appropriate opportunities to speak in naturalistic dialogue settings (Umair et al., 2024). Such results underscore the need to model intervention decisions—whether to speak, defer, clarify, or refuse—as explicit control actions rather than as byproducts of surface-level generation.

3.2 Definition and Core Classes of Frictive Intervention

We define a frictive intervention as follows.

Definition 1 (Frictive Intervention). *A frictive intervention is an intentional communicative act that temporarily resists immediate task completion in order to reduce epistemic, normative, or safety risk in an interaction.*

We distinguish a small set of core frictive intervention types that serve as the action space for Frictive Policy Optimization. These classes are not intended as an exhaustive theory of dialogue, but as a minimal set of epistemic control actions sufficient to capture the dominant alignment failure modes in current language models:

- **Clarification**, which seek missing constraints or resolve underspecification before acting.
- **Verification**, which check the validity of assumptions, premises, or sources.
- **Challenge**, which question or reject false or inconsistent presuppositions.
- **Redirection**, which reframe ill-posed tasks toward coherent alternatives.

- **Refusal**, which block actions that violate safety, ethical, or normative constraints.
- **Meta-dialogical interventions**, which explicitly manage the model’s own uncertainty or limitations.

Each class is defined functionally by its role in regulating epistemic commitment and risk, rather than by its surface linguistic form. In all cases, the defining property is that the intervention temporarily resists immediate task completion in order to improve downstream epistemic or normative outcomes.

This definition has three important components:

- **Intentionality**: The intervention is not noise or uncertainty leakage, but a deliberate control decision.
- **Resistance**: The intervention delays, redirects, or blocks immediate task execution.
- **Risk reduction**: The purpose of the intervention is to improve epistemic quality or prevent downstream failure.

Under this definition, hesitation without purpose, verbosity, or generic hedging do *not* count as friction interventions. Friction is not a matter of style, but of **control**.

A full formal taxonomy and fine-grained subtypes is presented in [Pustejovsky and Zhu \(2026\)](#), to which we refer the reader for detailed classification and annotation guidelines. In the present paper, we use these classes only to define the action space over which intervention policies are learned.

3.3 Frictive Interventions as Actions

A central claim of this paper is that frictive interventions must be treated as *actions* rather than stylistic variants of answers. This claim has three implications:

1. Interventions change the *state of the interaction*, not just its wording.
2. Interventions incur explicit costs (time, user effort, refusal penalties).
3. Interventions have delayed effects on epistemic risk.

These properties cannot be captured by token-level preferences or response re-ranking alone.

They require a control-theoretic treatment in which the model decides *whether* to act before deciding *how* to phrase the act.

The taxonomy introduced here defines the action space $\mathcal{A}$ used in subsequent sections. Each frictive intervention corresponds to a distinct action type that can be selected by a policy based on its expected epistemic consequences.

This move—from descriptive taxonomy to formal action space—marks the transition from conceptual framing to formal modeling. In the next section, we show how these intervention types are embedded in a risk-sensitive epistemic control framework.

4 Epistemic Friction as Risk-Sensitive Control

We model dialogue as a controlled stochastic process in which the agent regulates epistemic commitment under uncertainty. Let $h_t \in \mathcal{H}$ denote the observable interaction history up to time t , including the dialogue transcript, multimodal observations, task context, and any available external signals. We treat h_t as the sufficient interaction state on which policies, risk estimates, and friction functionals are conditioned. Let b_t denote the agent’s belief state at time t , summarizing its uncertainty over latent world states.

Belief Update. We assume beliefs evolve according to a Bayesian or learned update rule:

$$(1) \quad b_{t+1} = \text{BeliefUpdate}(b_t, h_{t+1})$$

This belief update equation places our framework within the classical theory of control under partial observability, in which the belief state b_t is treated as a sufficient statistic for the interaction history. In partially observable Markov decision processes (POMDPs), optimal control is performed over evolving belief distributions rather than over raw observations ([Åström, 1965](#); [Smallwood and Sondik, 1973](#); [Kaelbling et al., 1998](#)).

In dialogue systems, this formulation underlies probabilistic dialogue state tracking and grounding, where the agent maintains and updates an internal epistemic state in response to user observations ([Traum, 1994](#); [Clark, 1996](#); [Williams and Young, 2007](#)). More recent neural dialogue managers similarly operate by learning approximate belief update rules from interaction data ([Henderson et al., 2014](#); [Lee et al., 2019](#)). Our formulation makes this assumption explicit: alignment is a control problem

over latent epistemic states, not merely a mapping from histories to surface responses.

This equation defines the epistemic state transition: after observing the new dialogue history h_{t+1} , the agent updates its belief state from b_t to b_{t+1} . This makes explicit that alignment is a control problem over evolving beliefs, not just over surface responses.

Risk-Sensitive Objective. We define the agent’s objective as a discounted risk-sensitive return. Let y_t denote the surface realization (utterance) generated at time t . We assume that y_t is conditionally sampled from a realization policy $\pi_{\text{gen}}(y_t \mid h_t, a_t)$ after selecting an intervention type a_t , as formalized below in Eq. (2).

$$(2) \quad J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t \left(R_{\text{task}}(h_t, a_t, y_t) - \lambda C_{\text{fric}}(a_t) - \eta \mathcal{R}_{\text{risk}}(b_t, h_t, a_t, y_t) \right) \right]$$

This objective follows the standard discounted reinforcement learning formulation (Puterman, 1994; Kaelbling et al., 1998), augmented with an explicit risk-sensitive penalty as in classical risk-sensitive and constrained Markov decision processes (Howard and Matheson, 1972; Tamar et al., 2015; Chow et al., 2015; Achiam et al., 2017). In the service of FPO, this objective defines the central tradeoff of the model: the policy π maximizes discounted task reward while subtracting both an immediate intervention cost C_{fric} and an epistemic risk functional $\mathcal{R}_{\text{risk}}$. The weights λ and η control how strongly the agent prefers low-friction and low-risk trajectories relative to short-term task success.

The terms have the following interpretation:

- R_{task} measures immediate task utility from producing content.
- $C_{\text{fric}}(a_t)$ is the intervention cost, encoding user burden, latency, or refusal penalties.
- $\mathcal{R}_{\text{risk}}(b_t, h_t, a_t, y_t)$ measures expected downstream epistemic or normative failure under belief state b_t .

In what follows, when the belief state b_t is deterministically inferred from the observable history h_t , we suppress the explicit dependence on b_t and write $\mathcal{R}_{\text{risk}}(h_t, a_t, y_t)$ for notational simplicity.

Risk as Belief-Weighted Expectation. We model epistemic risk as an expectation over latent states:

$$(3) \quad \mathcal{R}_{\text{risk}}(b_t, h_t, a_t, y_t) = \mathbb{E}_{s \sim b_t} [r_{\text{risk}}(s, h_t, a_t, y_t)]$$

This is the standard belief-state cost formulation in POMDPs (Åström, 1965; Kaelbling et al., 1998; Young et al., 2013), here interpreted as epistemic and normative risk. Here we define epistemic risk as the belief-weighted expectation of a latent risk function r_{risk} . It makes explicit that risk depends on unobserved world states s , and that the agent must reason under epistemic uncertainty rather than optimizing a fully observed cost.

Policy Factorization. We factor the policy into intervention type and surface realization:

$$(4) \quad \pi(a_t, y_t \mid h_t) = \pi_{\text{int}}(a_t \mid h_t) \pi_{\text{gen}}(y_t \mid h_t, a_t)$$

This factorization mirrors hierarchical dialogue policies and option-based control, in which a high-level decision selects an intervention type and a conditional generator realizes it (Williams and Young, 2007; Young et al., 2013; Sutton et al., 1999; Bacon et al., 2017). In this formulation, it separates the decision of *what kind of epistemic intervention* to perform (answer, clarify, refuse, etc.) from the decision of *how to realize it linguistically*. This is crucial for modeling intervention as a first-class control action.

Clarification as Optimal Epistemic Control

The FPO framework treats intervention type as a first-class decision variable, but this raises a natural normative question: under what conditions is clarification actually the optimal action? If intervention costs always dominate, the framework collapses to standard response generation; if clarification is always preferred, it degenerates into pathological hesitation. We therefore establish a simple dominance result showing that clarification is optimal precisely when its expected epistemic benefit outweighs its interaction cost. This yields an explicit threshold criterion that the FPO methods are designed to learn to approximate from data.

Proposition 1 (Threshold Optimality of Clarification). *Consider a decision point at time t with belief state b_t and history h_t . Let a_c denote a clarificatory action and a_a denote a direct answer action. Suppose:*